

VULNERABILITY MANAGEMENT REPORT

From Detection to Decision: Finding, Prioritizing, and Fixing Real Vulnerabilities

Project Detail	Information
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Project	Vulnerability Management Sprint
Environment	Wazuh Homelab
Assessment Period	September 2026

1. Executive Summary

This project focused on vulnerability management using an existing Wazuh homelab. The goal was to move beyond simply detecting vulnerabilities and instead review, prioritize, and take action on the findings.

The assessment covered two endpoints: a Windows 11 endpoint and an Ubuntu/Linux endpoint. Wazuh Vulnerability Detection and Syscollector were used to collect endpoint information and identify vulnerabilities across both systems.

The vulnerability findings were reviewed and prioritized using three main factors: CVSS severity, CISA Known Exploited Vulnerabilities (KEV) status, and asset criticality. This was done to avoid treating every vulnerability the same way based only on severity.

Severity / Status	Number of Findings
Critical	17
High	445
Medium	284
Low	52
Pending Evaluation	289
Total	1,087

The highest-priority finding was CVE-2026-33824, affecting the Windows IKE Extension (IKEEXT). It was classified as P1 – Immediate because it was Critical, listed in the CISA KEV catalog, and affected the domain-facing Windows endpoint, which was the highest-criticality asset in the environment.

As part of the remediation phase, the IKEEXT service was stopped and disabled using PowerShell. This reduced the attack surface, but Wazuh continued to report the vulnerability because the underlying component had not been patched. This demonstrated the difference between mitigating a vulnerability and completely remediating it.

The project also identified 34 Linux findings that appeared to require further verification because older CVEs were being reported against supported Ubuntu packages. These were kept in the P3 backlog rather than being immediately removed from the inventory.

2. Scope & Objective

2.1 Scope

Area	Scope
SIEM / Vulnerability Platform	Wazuh
Windows Endpoint	Hayzed-Windows-VM
Linux Endpoint	hayzed
Windows OS	Windows 11 Pro
Linux OS	Ubuntu 26.04

Vulnerability Detection	Wazuh Vulnerability Detection
Endpoint Inventory	Wazuh Syscollector
Prioritization	CVSS, CISA KEV and Asset Criticality
Remediation	IKEEXT mitigation

2.2 Asset Context

Asset	Operating System	Role / Criticality
Hayzed-Windows-VM	Windows 11 Pro	Domain-facing endpoint – High criticality
hayzed	Ubuntu 26.04	Internal server – Medium criticality

2.3 Objectives

1. Enable and configure Wazuh Vulnerability Detection.
2. Confirm endpoint inventory information was being collected through Syscollector.
3. Build a vulnerability inventory across the Windows and Linux endpoints.
4. Cross-reference relevant vulnerabilities against the CISA KEV catalog.
5. Prioritize vulnerabilities using CVSS, KEV status, and asset criticality.
6. Identify findings that may require additional verification.
7. Remediate or mitigate at least one significant vulnerability.
8. Validate the action using Wazuh.
9. Document the vulnerabilities found, their priorities, the action taken, and the remaining risks.

3. Methodology

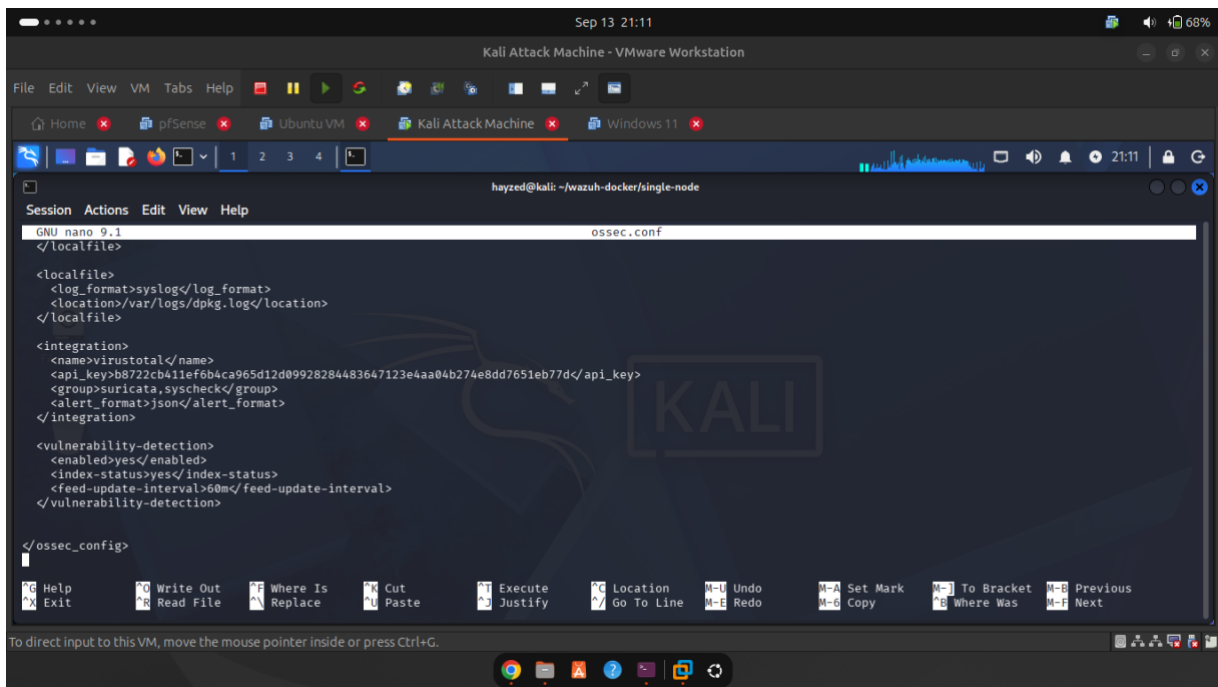
The project was completed through four main stages: Detection Setup; Vulnerability Discovery and Inventory; Vulnerability Triage and Prioritization; and Remediation and Validation.

3.1 Detection Setup

Wazuh Vulnerability Detection was enabled on the Wazuh manager.

```
<vulnerability-detection>
  <enabled>yes</enabled>
  <index-status>yes</index-status>
  <feed-update-interval>60m</feed-update-interval>
</vulnerability-detection>
```

This enabled the vulnerability detection module, enabled vulnerability indexing, and configured the vulnerability feed to update every 60 minutes.

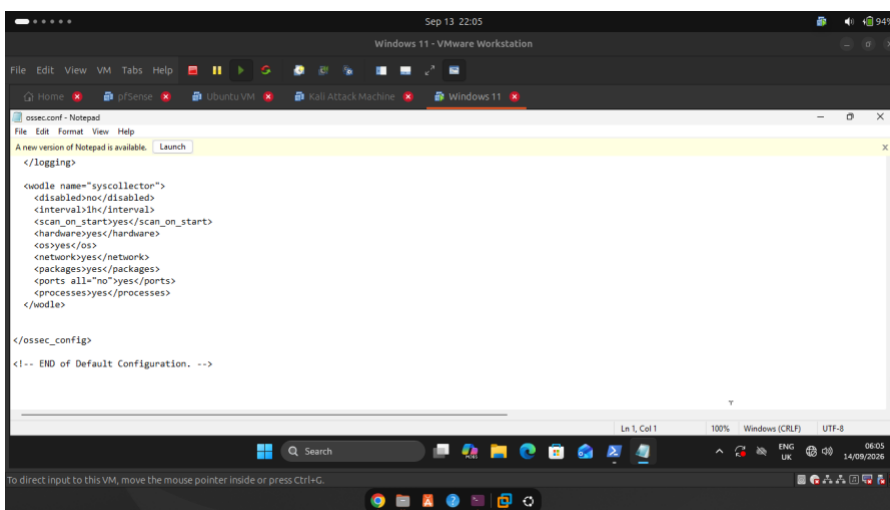


3.2 Syscollector Configuration

Wazuh Syscollector was enabled on both endpoints to collect information about the systems and installed software.

```
<wodle name="syscollector">
  <disabled>no</disabled>
  <interval>1h</interval>
  <scan_on_start>yes</scan_on_start>
  <hardware>yes</hardware>
  <os>yes</os>
  <network>yes</network>
  <packages>yes</packages>
  <ports>no</ports>
  <processes>yes</processes>
</wodle>
```

The Linux endpoint used the same Syscollector configuration.



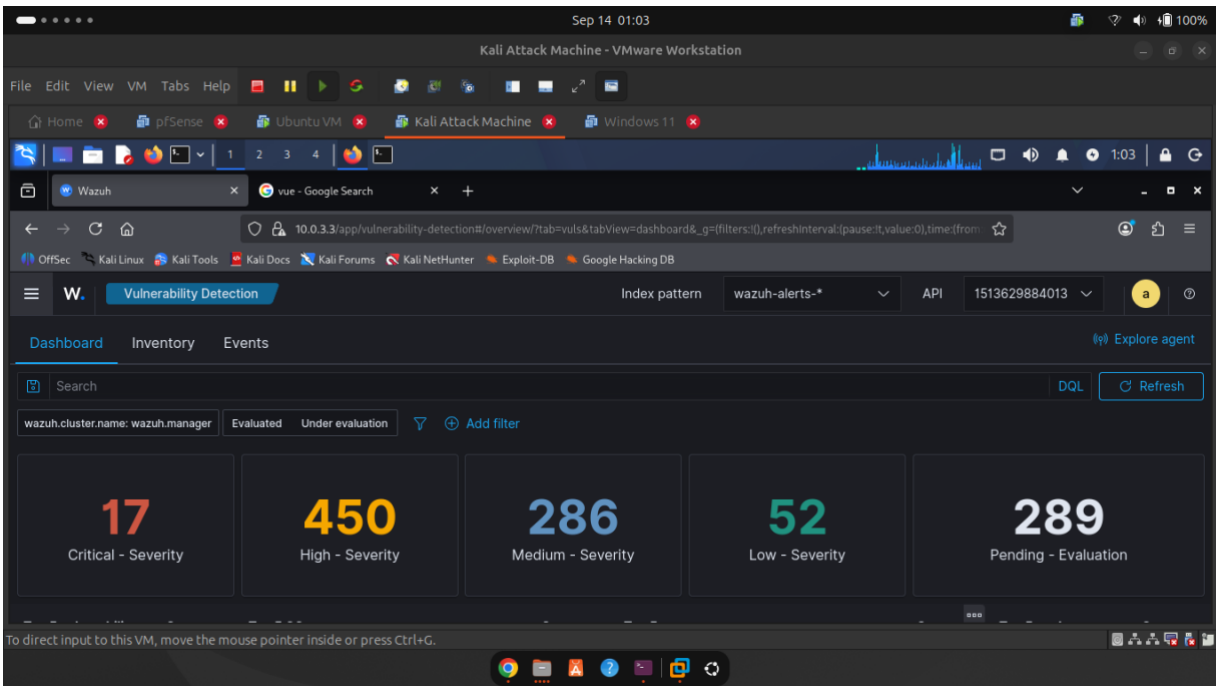
3.3Vulnerability Discovery and Inventory

After Vulnerability Detection and Syscollector were configured, the Wazuh Vulnerability Detection dashboard was reviewed. The vulnerability inventory was exported separately for the Windows and Linux endpoints. The Windows export contained 283 findings, while the Linux export contained 804 findings.

Severity / Status	Windows	Linux	Total
Critical	7	10	17
High	218	227	445
Severity / Status	Windows	Linux	Total
Medium	57	227	284
Low	1	51	52
Pending Evaluation	0	289	289
Total	283	804	1,087

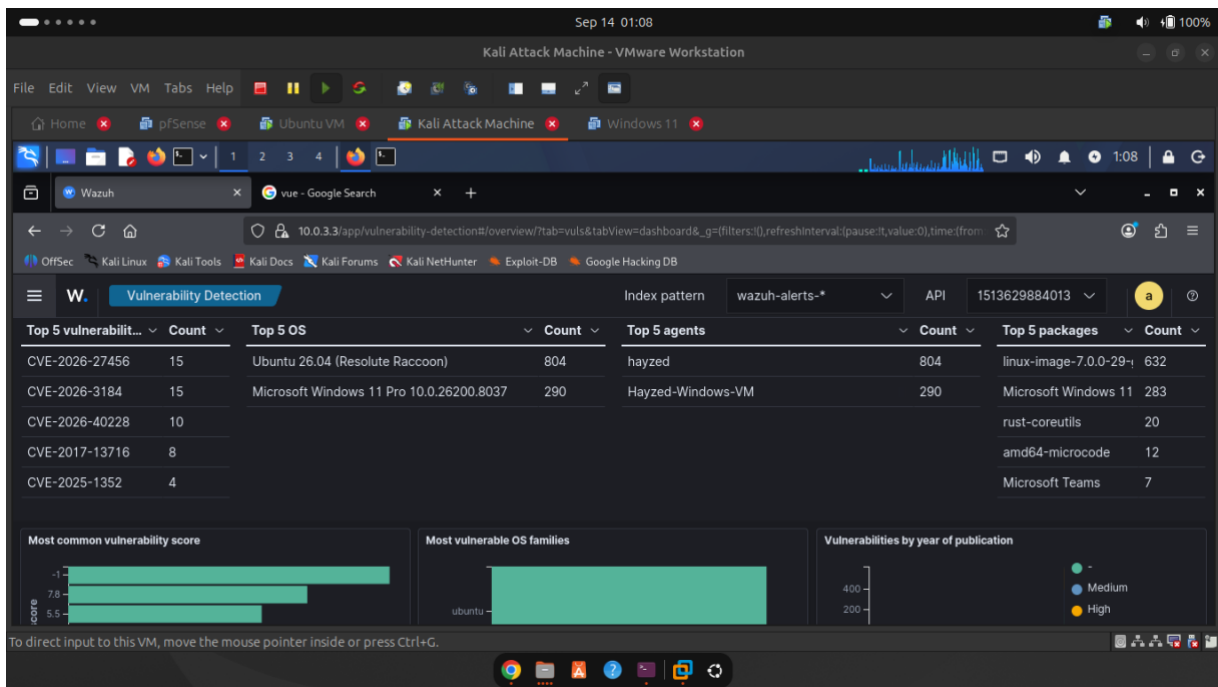
3.4Wazuh Vulnerability Dashboard

The Wazuh dashboard provided an overview of the vulnerabilities detected in the environment.



The dashboard snapshot contained 1,094 findings, while the exported CSV files used for the prioritization worksheet contained 1,087 findings. The difference is associated with the fact that the dashboard and exported data were captured at different points in time. The 1,087 findings in the CSV exports and Excel worksheet were used as the dataset for final prioritization.

3.5 Vulnerability Overview



The dashboard showed Ubuntu 26.04 and Microsoft Windows 11 Pro among the affected operating systems. The package section showed Linux kernel packages, Microsoft Windows, rust-coreutils, amd64-microcode, and Microsoft Teams among the packages with vulnerability findings.

4. Findings

4.1 Vulnerability Prioritization

The vulnerability inventory contained a large number of Critical and High findings. However, severity alone was not used to determine which vulnerability should be handled first.

The prioritization process considered CVSS + CISA KEV + Asset Criticality. CVSS was used to understand the technical severity of the vulnerability. CISA KEV status was used to identify vulnerabilities that are known to have been exploited. Asset criticality was used to determine the importance of the affected endpoint within the environment.

The Windows endpoint was considered the highest-criticality asset because it was the domain-facing endpoint.

4.2 Prioritization Results

Priority	Number of Findings
P1 – Immediate	1
P2 – This Week	461
P2 – This Week, Elevated: KEV	1
P3 – Backlog	335
Pending Evaluation	289
Total	1,087

The prioritization resulted in one P1 finding, a large P2 workload, a P3 backlog, and a group of findings that were still pending evaluation.

4.3P1 Finding – CVE-2026-33824

Attribute	Result
CVE	CVE-2026-33824
Affected Component	Windows IKE Extension (IKEEXT)
Severity	Critical
CVSS Range	9.0–10.0
CISA KEV	Yes
Affected Asset	Hayzed-Windows-VM
Asset Criticality	High – Domain-facing endpoint
Priority	P1 – Immediate

P1 Justification: CVE-2026-33824 is prioritized as P1 because it carries a Critical CVSS score, is confirmed in the CISA KEV catalog as actively exploited in the wild, and affects the domain-facing Windows endpoint, the highest criticality asset in this environment.

4.4Additional KEV Finding

Attribute	Result
CVE	CVE-2026-32202
Severity	Medium
CISA KEV	Yes
Affected Asset	Hayzed-Windows-VM
Priority	P2 – This Week, Elevated

This finding was placed in the elevated P2 category because it was listed in the CISA KEV catalog.

4.5P2 Findings

CVE	Affected Area
CVE-2026-41096	Windows DNS
CVE-2026-42904	Windows TCP/IP
CVE-2026-44815	Windows DHCP Client
CVE-2026-45657	Windows Kernel
CVE-2026-45602	Windows DHCP Server

A total of 461 findings were classified as standard P2 – This Week. The Linux environment also contained Critical and High findings that were considered during the prioritization process.

4.6 P3 Findings

A total of 335 findings were placed in the P3 backlog. These findings were considered lower priority compared with the P1 and P2 findings or required additional verification before a final decision could be made. The purpose of the P3 category was to maintain visibility of the findings without treating all vulnerabilities as immediate incidents.

4.7 Suspected False Positives

During the review of the Linux findings, 34 findings were identified as a likely false-positive pattern. These findings involved older CVEs being reported against supported Ubuntu packages. One example visible in the Wazuh dashboard was CVE-2017-13716.

The findings were not immediately removed from the inventory. Instead, they were retained in the P3 backlog pending manual verification against the installed Ubuntu package versions and relevant package changelogs.

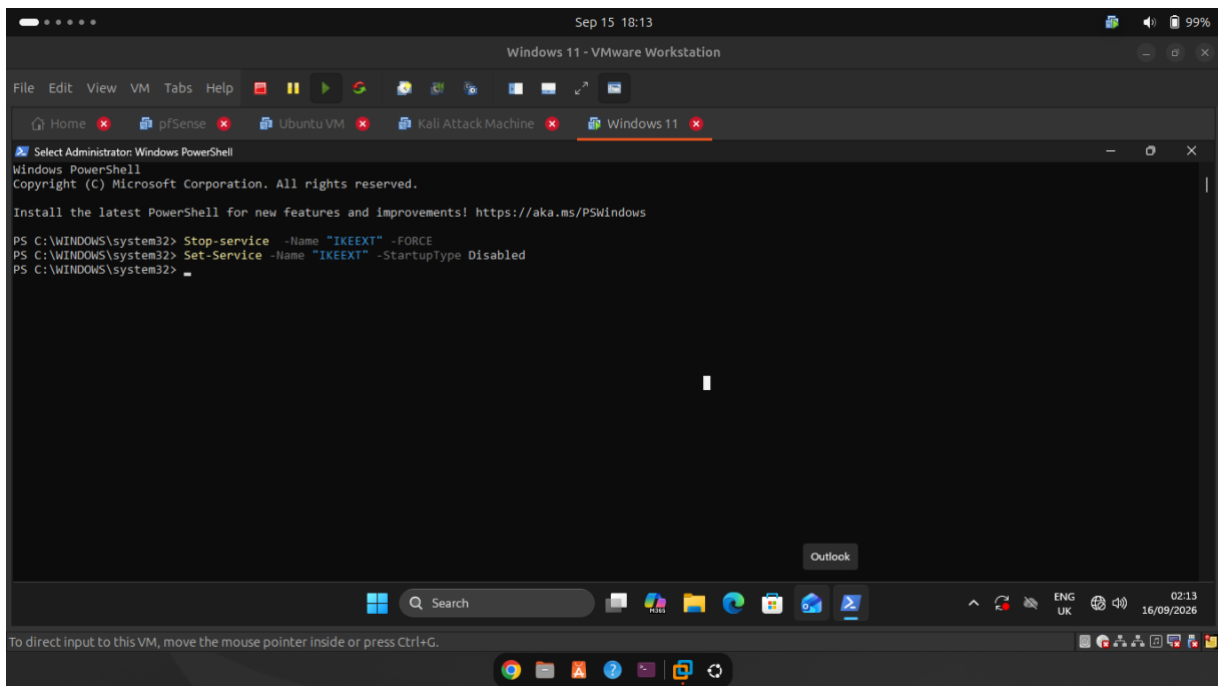
5. Remediation

5.1 Selected Vulnerability

CVE-2026-33824 – Windows IKE Extension (IKEEXT). Before the mitigation, the IKEEXT service was active and enabled on the Windows endpoint. The objective was to reduce the exposure of the affected service while full patching was still pending.

5.2 IKEEXT Mitigation

```
Stop-Service -Name "IKEEXT" -Force  
Set-Service -Name "IKEEXT" -StartupType Disabled
```



The first command stopped the IKEEXT service. The second command changed the service startup type to Disabled. The commands completed without an error. This reduced the attack surface associated with the affected service.

5.3 Validation

Action	Result
IKEEXT service stopped	Completed
IKEEXT startup disabled	Completed
Attack surface reduced	Yes
Underlying vulnerability patched	No
Wazuh vulnerability cleared	No
Full remediation	Pending

After the mitigation, the vulnerability was reviewed again through Wazuh. Wazuh continued to report CVE-2026-33824. This was expected because disabling the IKEEXT service did not patch or remove the underlying vulnerable component.

The mitigation therefore reduced the risk but did not constitute full remediation. The next step is to apply the appropriate vendor patch/system update and perform another Wazuh scan to confirm whether the vulnerability has been cleared.

6. Gaps and Challenges

6.1 Wazuh CVE Feed Issue

One of the challenges encountered during the project was that the Wazuh deployment running in Docker was initially unable to download the CVE feed. The issue was resolved by adding DNS configuration to the Docker Compose configuration. After the DNS configuration was corrected, Wazuh was able to continue with vulnerability feed processing.

6.2 Slow Network Connection

Another challenge was the slow network connection during the remediation stage. Because of the slow connection, the required vendor patch could not be completed within the project period. As a result, the IKEEXT service was disabled as a mitigation instead of completing the full patching process.

6.3 Pending Evaluation Findings

The exported dataset contained 289 findings that were still pending evaluation. These findings were kept separate from the evaluated vulnerability priorities because their final severity had not yet been established.

7. Limitations

The project was conducted in a homelab environment containing only two endpoints. Therefore, the results represent the vulnerability state of this environment and do not represent a large enterprise network.

Another limitation was the difference between the Wazuh dashboard snapshot and the exported vulnerability data. The dashboard showed 1,094 findings while the CSV and Excel dataset contained 1,087 findings. The exported dataset was used for the final prioritization.

The IKEEXT vulnerability could also not be fully patched during the project because of the network limitation. The service was mitigated by stopping and disabling it, but the underlying vulnerability remained present.

Finally, the 34 suspected Linux false positives still require manual package and changelog verification before they can be confirmed as false positives.

8. Recommendations and Next Steps

8.1 Complete Vendor Patching

The Windows endpoint should be updated with the appropriate security patches once the network issue has been resolved. CVE-2026-33824 should remain tracked until Wazuh confirms that the vulnerability has been cleared.

8.2 Re-scan After Patching

After patching, the Windows endpoint should be rescanned through Wazuh. The purpose of the rescan is to confirm whether the vulnerability has been removed rather than relying only on the service being disabled.

8.3 Review P2 Findings

The P2 findings should continue through the vulnerability management process according to their priority, affected assets, and available remediation options.

8.4 Review Pending Findings

The 289 pending evaluation findings should be reviewed after their vulnerability information becomes available. They should then be assigned an appropriate severity and priority.

8.5 Verify Suspected False Positives

The 34 suspected Linux false positives should be manually verified using the installed package versions and relevant Ubuntu package information before being removed from the backlog.

8.6 Continue Vulnerability Monitoring

Wazuh Vulnerability Detection and Syscollector should remain enabled so that changes in installed software and newly disclosed vulnerabilities can continue to be identified.

9. Conclusion

This project showed that vulnerability management is not simply about finding vulnerabilities. The larger task is understanding the findings and deciding what needs attention first.

The Wazuh assessment identified 1,087 exported vulnerability findings across the Windows and Linux endpoints. These findings were reviewed using CVSS severity, CISA KEV status, and asset criticality.

The main P1 finding was CVE-2026-33824, affecting the Windows IKE Extension (IKEEXT). It was prioritized as P1 because it was Critical, listed in CISA KEV, and affected the domain-facing Windows endpoint.

The finding was mitigated by stopping and disabling the IKEEXT service. Wazuh continued to report the vulnerability after the mitigation, demonstrating that disabling an affected service can reduce the attack surface without removing the underlying vulnerability.

The main lesson from the project was that mitigation is not the same as patching. A mitigation can reduce risk while a permanent fix is still pending, but the vulnerability should remain tracked until the underlying issue has been addressed and validated.

10. Appendix

Appendix A – Wazuh Vulnerability Detection Configuration

```
<vulnerability-detection>
  <enabled>yes</enabled>
  <index-status>yes</index-status>
  <feed-update-interval>60m</feed-update-interval>
</vulnerability-detection>
```

Appendix B – Syscollector Configuration

```
<wodle name="syscollector">
<disabled>no</disabled>
  <interval>1h</interval>
  <scan_on_start>yes</scan_on_start>
<hardware>yes</hardware>
  <os>yes</os>
  <network>yes</network>
  <packages>yes</packages>
<ports>no</ports>
  <processes>yes</processes>
</wodle>
```

Appendix C – IKEEXT Mitigation Commands

```
Stop-Service -Name "IKEEXT" -Force
Set-Service -Name "IKEEXT" -StartupType Disabled
```

Appendix D – Vulnerability Summary

Item	Result
Windows Findings	283
Linux Findings	804
Total Exported Findings	1,087
Critical	17
High	445
Medium	284
Low	52
Pending Evaluation	289
P1 – Immediate	1
P2 – This Week	461
P2 – Elevated KEV	1
P3 – Backlog	335

Suspected Linux False Positives	34
KEV Findings	2
Main P1 Finding	CVE-2026-33824
Mitigation	IKEEXT stopped and disabled